

FORESIGHT — Executive Readout

Demand Forecasting & Inventory Risk | NorthBay Living

The project turns transaction history and inventory information into a weekly planning view: what demand is likely to look like, where stockout or overstock exposure is visible, and which SKUs need attention first.

Decision layer	Current output
Forecast	Weekly SKU-level demand; 8-week forward forecast produced
Risk	Stockout / overstock flags with action and rupee impact
Dashboard	Actual vs forecast, risk overview and priority actions
Service	FastAPI scoring service with single-SKU and batch endpoints

Primary actions: Reorder Now, Markdown/Clear, and Healthy.

1. The financial picture

The current risk-scored dataset contains 8,657 SKU-week records. Its aggregate exposure is useful as a planning signal, but it should not be read as a guaranteed loss or a forecast of realised financial impact.

Measure	Current calculated exposure
Sales at risk from stockout flags	■11,374,843
Locked capital from overstock	■38,307,979
Combined value at stake	■49,682,822
Stockout units at risk	18,175.1 units
Overstock units at risk	88,674.2 units

These values are calculated from the forecast, stock position, selling price and cost fields in the risk dataset. They are decision-support numbers and should be validated by Operations/Finance before being treated as a financial commitment.

2. What the planner should do next

Action	Meaning in the current implementation	Operational use
Reorder Now	Forecast creates a stockout gap	Review highest-value items first; confirm stock and lead time
Markdown/Clear	Excess stock is above the risk threshold	Review price, promotion and clearance options
Healthy	No immediate stockout/overstock flag	Continue normal monitoring

The system does not place purchase orders or make pricing changes automatically. It produces a prioritised view so a planner can make the final call with current operational context.

3. What is already available

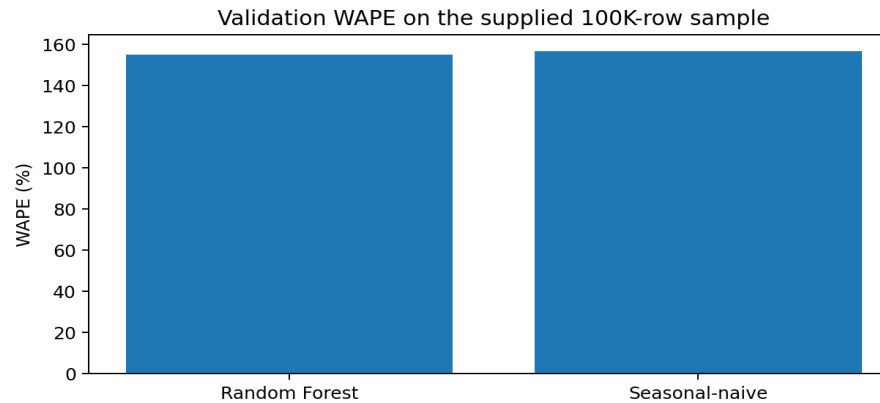
The dashboard is designed as the point where the analysis becomes usable by a non-technical planner.

Dashboard view	Purpose
Category / SKU filters	Narrow the view to the product set being reviewed
Actual vs Forecast	Compare observed demand with model demand by week
Risk Overview	See stockout, overstock and healthy counts
Priority Actions	Surface the reorder and markdown workload
Action Summary	Give a compact count of each decision category

The live dashboard is the operational front end; the API is the programmatic route for SKU-level scoring.

4. Forecast performance — keep the result honest

Metric	Result
Model test WAPE	155.05%
Seasonal-naive WAPE	156.65%
Difference	1.60 percentage points
Rolling-origin mean WAPE	158.68%
Leakage audit	PASS
Forward horizon	8 weeks



Interpretation: the model improves on the seasonal-naive reference on the reported same-row test comparison, but the improvement is modest and the absolute WAPE remains high. The result should therefore be treated as a planning signal that needs monitoring, not as a high-accuracy production forecast.

5. What the forecast result means in practice

The strongest part of the current solution is the chain from forecast to decision, not a claim of perfect forecast accuracy.

Step	Current implementation
Historical demand	Weekly SKU demand is created from cleaned transactions
Reference	Seasonal-naive baseline is calculated
Model	Random Forest is trained on engineered lag / rolling / calendar features
Validation	Chronological test plus rolling-origin validation
Decision layer	Forecast is compared with stock position to create risk gaps
Action	Risk is converted to Reorder Now, Markdown/Clear or Healthy

For operational use, the forecast should be read together with stock on hand, reorder point, safety stock, lead time and current merchandising context.

6. Assumptions and limitations

A few points should remain visible in any stakeholder handoff:

- The executed model validation uses the supplied 100K transaction sample, while the original source is larger. Results can change when a broader history is processed.
- WAPE is high on this sample. A small percentage-point improvement over baseline does not make the forecast highly accurate by itself.
- Risk values depend on the current forecast, inventory position, selling price and cost inputs.
- Sparse or newly introduced SKUs have less historical evidence and can be harder to forecast reliably.
- Promotions and other external demand drivers should be checked before a high-impact operational decision is finalised.
- **The solution is decision support. It does not automate purchase-order placement or replace the planner's final judgement.**

7. Recommended operating rhythm

When	Action
At each forecast refresh	Review the highest-value Reorder Now and Markdown/Clear records
Before ordering	Confirm physical stock, open orders and supplier lead time
Before clearance	Check merchandising context, price and promotion plans
Monthly / regular model review	Track WAPE and bias; compare with the seasonal-naive reference
When the full history is available	Re-run the pipeline and refresh the forward forecast

Current risk dataset workload: 8,064 Reorder Now records, 503 Markdown/Clear records and 90 Healthy records.

These are record counts in the current risk output, not counts of unique products.

8. Delivery and access

The stakeholder-facing pieces are in place: a live dashboard, a deployed scoring API, the reproducible analysis/model code, and the project readout materials.

Resource	Address
Live dashboard	https://foresight-project-cs3i.onrender.com
Live scoring service	https://foresight-scoring-api-6nec.onrender.com
API documentation	https://foresight-scoring-api-6nec.onrender.com/docs
Repository	https://github.com/hrutikchavan-ai/FORESIGHT-PROJECT

Closing note

FORESIGHT is now structured as a complete analytics workflow: cleaned data → weekly demand → forecast → risk decision → dashboard/API. The next focus should be reproducibility and repository handoff, followed by final validation of the live deployment.

Prepared for project submission | Demand & Inventory Intelligence